

Super-Eddington Little Blue Dots Over-Reionize Helium

JWST's faint AGN may finish helium reionization too early unless only a small fraction are super-Eddington.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

arXiv:2608.28745 doi:10.48550/arXiv.2608.28745



Background

JWST found many faint broad-line AGN at $z > 4$, dubbed Little Blue Dots. Their blue continua and compact sizes suggest super-Eddington accretion with strong extreme-UV but weak X-ray emission.

Do these super-Eddington LBDs reionize helium too early, conflicting with observations that He II reionization ended near $z \approx 3$?

What this paper is selling

The paper tests whether super-Eddington LBDs can reionize helium without violating timing constraints. It finds that only a small fraction of LBDs can be super-Eddington, unless their escape fractions are tiny.

Key results to remember

- He II reionization ending at $z \approx 3$ limits super-Eddington LBDs to $\lesssim 10\%$ of the population.
- A majority super-Eddington scenario requires escape fractions $\lesssim 1.5\%$, below observed values.
- Constraints depend on the extreme-UV spectrum shape of super-Eddington black holes.

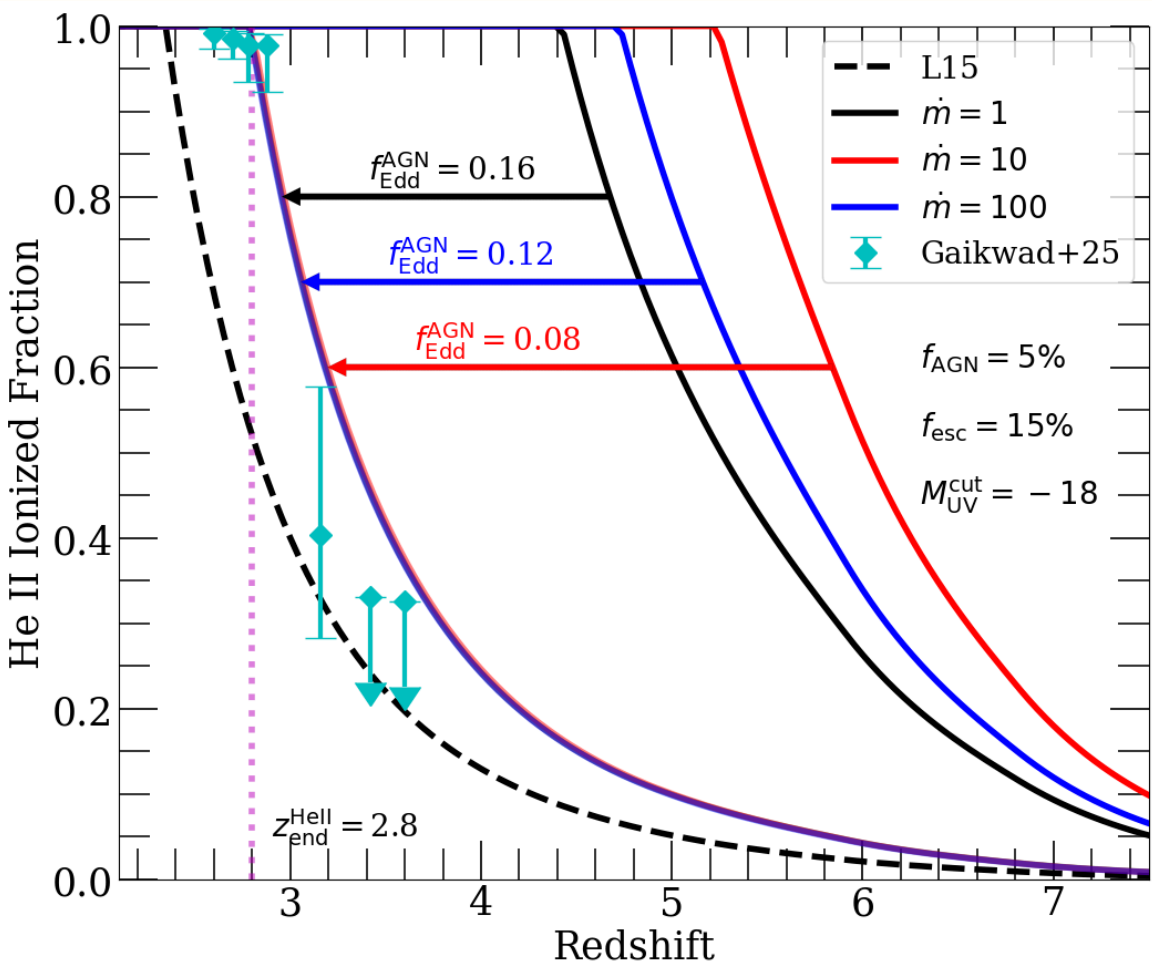


Figure 3 | He II reionization histories show that super-Eddington models finish too early unless the AGN fraction is small.

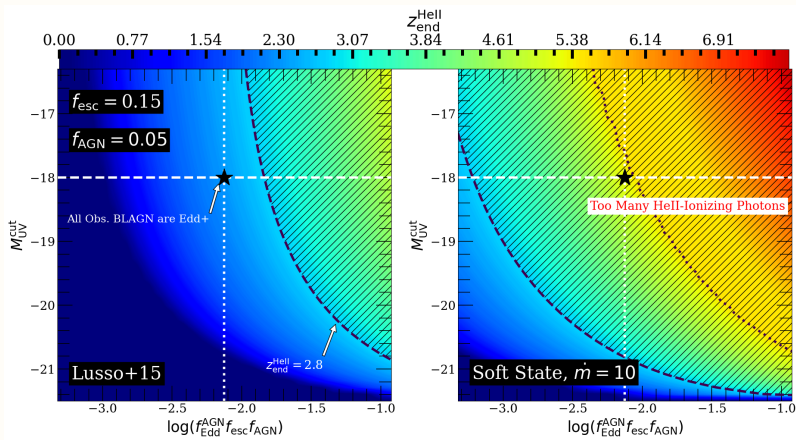


Figure 4 | Excluded parameter space: the hatched region marks super-Eddington fractions that violate He II reionization timing.